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(57) Abstract:

The present invention relates to an artificial intelligence-based system for detecting and responding to spoofing attacks in real-time network environments. The system comprises a data ingestion module configured to collect network traffic data from multiple devices and extract feature vectors including network-level, device-level, and behavioral attributes. A behavioral pattern learning engine employing a bidirectional long short-term memory (BiLSTM) network analyzes temporal sequences of network events to generate a behavioral deviation score indicative of spoofing activity. A real-time anomaly detection engine utilizes an autoencoder model and an isolation forest model to identify abnormal patterns within the extracted features. A predictive threat intelligence engine incorporating a long short-term memory (LSTM) predictor estimates the probability and category of potential upcoming attacks based on historical traffic sequences. An adaptive policy network using a deep Q-network (DQN) dynamically adjusts detection thresholds and selects appropriate response actions according to current network conditions. A multi-layer correlation engine aggregates outputs from multiple analytical layers to generate a spoofing decision and trigger automated real-time mitigation actions. The invention provides an adaptive and intelligent framework for accurate detection, prediction, and response to spoofing attacks, thereby enhancing overall network security. Fig 1 to 11.

FORM 2

THE PATENTS ACT, 1970

(39 of 1970)

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THE PATENTS RULES, 2003

COMPLETE SPECIFICATION

(See Section 10; rule 13)

Title of the Invention

**“An AI-based multi-layer spoofing detection and response system
for real-time network security”**

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- 5 The following specification particularly describes the invention and the manner in which it is performed.

TECHNICAL FIELD

[0001] The present invention relates to the field of network security and intrusion detection systems. The invention relates to an artificial intelligence-based multi-layer spoofing detection and response system that utilizes machine
5 learning and deep learning techniques for real-time detection, prediction, and mitigation of spoofing attacks in computer networks.

BACKGROUND ART

[0002] Background description includes information that may be useful in understanding the present invention. It is not an admission that any of the
10 information provided herein is prior art or relevant to the presently claimed invention, or that any publication specifically or implicitly referenced is prior art.

[0003] With the rapid expansion of digital communication networks, the number of connected devices and network services has increased significantly. Modern network infrastructures support large volumes of data exchange among
15 computers, mobile devices, Internet of Things (IoT) devices, and cloud-based services. While this connectivity enhances communication and operational efficiency, it also exposes networks to various security threats, including spoofing attacks.

[0004] Spoofing attacks occur when a malicious entity impersonates a
20 legitimate device, user, or network source by falsifying identifying information such as Internet Protocol (IP) addresses, media access control (MAC) addresses, or other network identifiers. Such attacks enable adversaries to bypass authentication mechanisms, gain unauthorized access to network resources, intercept sensitive data, or disrupt normal network operations. Common forms of
25 spoofing include IP spoofing, address resolution protocol (ARP) spoofing, DNS spoofing, and device identity spoofing.

[0005] Conventional network security mechanisms, such as firewalls, rule-based intrusion detection systems, and signature-based security tools, often rely on predefined attack signatures or static threshold rules. Although these
30 approaches can detect known attack patterns, they are generally ineffective in identifying sophisticated or evolving spoofing techniques. Attackers frequently

modify their behavior to evade signature-based detection systems, making it difficult for traditional systems to accurately identify spoofing attempts in real time.

[0006] Recent advances in artificial intelligence and machine learning have introduced new opportunities for improving network security. Techniques such as deep learning, anomaly detection models, and reinforcement learning can analyze complex patterns in large volumes of network traffic data. However, many existing AI-based solutions focus on isolated detection mechanisms and do not integrate behavioral modeling, anomaly detection, predictive intelligence, and adaptive response strategies within a unified architecture.

[0007] US12021895B2 A multi-level, ensemble network monitoring system for detection of suspicious network activity from one or more a plurality of user computing devices on an external network communicatively connected via a network server to a private communication network is disclosed. In malware detection, the ensemble network monitoring system comprises artificial intelligence (AI) with bidirectional long short-term memory (BDLSTM) recurrent neural networks (RNNs) and natural language processing (NLP) to predict possible security threats and then initiate remedial measures accordingly. Enabling a proactive approach to detection and prevention of potential malicious activity, the BDLSTM RNN may perform real-time monitoring and proactively forecast network security violations to block network communications associated with high-risk user computing devices from accessing a private communication network.

[0008] US12381891B2 A system includes a compute engine that implements artificial intelligence-based dynamic, predictive, statistical classification meta-algorithms for better information security threat identification. The closed loop output of this algorithm uses a dynamic weights model along with AI/ML/DL to identify diverse information security threats more comprehensively and more efficiently. This system evolves with time and self-corrects to adapt to the ever changing needs of cyber security.

[0009] There exists a need for an improved network security system capable

of accurately detecting spoofing attacks in real time while adapting to evolving attack strategies. A system that combines multiple artificial intelligence models, performs behavioral analysis over temporal network data, predicts potential threats, and dynamically responds to suspicious activities would significantly
5 enhance the reliability and effectiveness of network security mechanisms.

[0010] The present invention fulfils these limitations by providing an AI-based multi-layer spoofing detection and response system that integrates behavioral pattern learning, anomaly detection, predictive threat intelligence, and adaptive policy mechanisms to detect and mitigate spoofing attacks efficiently in
10 real time.

OBJECTS OF THE INVENTION

[0011] The principal object of the present invention is to overcome the disadvantages of the prior art.

[0012] The primary objective of the present invention is to provide an
15 artificial intelligence-based multi-layer spoofing detection and response system capable of identifying and mitigating spoofing attacks in real-time network environments.

[0013] An objective of the present invention is to provide a system that collects and processes network traffic data from multiple devices and extracts
20 network-level, device-level, and behavioral features for comprehensive traffic analysis.

[0014] Another objective of the present invention is to provide a behavioral pattern learning mechanism using a bidirectional long short-term memory (BiLSTM) network to analyze temporal sequences of network events and detect
25 deviations in device behavior indicative of spoofing activity.

[0015] Another objective of the present invention is to provide a real-time anomaly detection mechanism using an autoencoder model and an isolation forest model to identify abnormal network traffic patterns and statistical outliers.

[0016] Another objective of the present invention is to provide a predictive
30 threat intelligence engine capable of forecasting potential spoofing attacks and

identifying attack categories based on historical network traffic patterns using a long short-term memory (LSTM) predictor.

[0017] Another objective of the present invention is to provide an adaptive policy network based on a deep Q-network (DQN) reinforcement learning model
5 to dynamically adjust detection thresholds and automatically determine appropriate response actions according to changing network conditions.

[0018] Another objective of the present invention is to provide a multi-layer correlation engine configured to aggregate outputs from multiple detection and prediction modules in order to generate a reliable spoofing detection decision.

10 [0019] Another objective of the present invention is to enable automated real-time response actions such as alert generation, traffic filtering, device isolation, or network access control upon detection of spoofing activity.

[0020] Another objective of the present invention is to improve detection accuracy while reducing false positives and false negatives by combining multiple
15 artificial intelligence models within a unified framework.

[0021] Another objective of the present invention is to provide a scalable and adaptive network security system capable of protecting modern network infrastructures from evolving spoofing threats.

SUMMARY

20 [0022] The present invention relates to a system and method for detecting and responding to spoofing attacks in real-time network environments using artificial intelligence and multi-layer analytical techniques. The invention provides an intelligent network security framework capable of identifying abnormal behavior, predicting potential spoofing attempts, and automatically initiating appropriate
25 response actions.

[0023] The invention provides an AI-based multi-layer spoofing detection and response system comprising a data ingestion module configured to receive network traffic data from a plurality of devices and extract feature vectors containing network-level, device-level, and behavioral attributes. The extracted

feature vectors are processed by a behavioral pattern learning engine that utilizes a bidirectional long short-term memory (BiLSTM) network to analyze temporal sequences of network events and generate a behavioral deviation score indicative of potential spoofing behavior.

5 **[0024]** The system includes a real-time anomaly detection engine configured to identify abnormal traffic patterns using a combination of machine learning techniques. The anomaly detection engine employs an autoencoder model to compute reconstruction error from the feature vectors and an isolation forest model to generate statistical outlier scores, thereby improving the ability to detect
10 irregular network activity.

[0025] The invention includes a predictive threat intelligence engine configured to forecast possible upcoming attacks based on historical network behavior. The predictive engine employs a long short-term memory (LSTM) predictor to estimate the probability and category of potential attacks, enabling
15 proactive security measures.

[0026] The system comprises an adaptive policy network that utilizes a deep Q-network (DQN) reinforcement learning model to dynamically adjust detection thresholds and select appropriate response actions based on real-time network conditions and system state metrics.

20 **[0027]** The system includes a multi-layer correlation engine configured to aggregate outputs from the behavioral pattern learning engine, anomaly detection engine, predictive threat intelligence engine, and adaptive policy network. The correlation engine utilizes a machine learning classifier and confidence-weighted decision mechanisms to generate a final spoofing detection decision and trigger an
25 automated response in real time.

[0028] The invention provides a method for detecting spoofing attacks in a network by ingesting network traffic data, extracting relevant feature vectors, analyzing behavioral sequences using a BiLSTM network, detecting anomalies using autoencoder and isolation forest models, predicting future attacks using an

LSTM predictor, selecting adaptive response actions through a deep Q-network, and correlating outputs from multiple analytical layers to produce a spoofing detection decision and execute real-time mitigation actions.

5 [0029] The invention improves network security by combining behavioral analysis, anomaly detection, predictive intelligence, and adaptive policy learning within a unified multi-layer architecture, thereby enhancing spoofing detection accuracy, reducing false alarms, and enabling proactive and automated network defense mechanisms.

10 [0030] These and other features will become apparent from the following detailed description of illustrative embodiments thereof, which is to be read in connection with the accompanying drawings. While the invention has been described and shown with reference to the preferred embodiment, it will be apparent that variations might be possible that would fall within the scope of the present invention.

15 **BRIEF DESCRIPTION OF DRAWINGS**

[0031] The accompanying illustrations are incorporated into and form a part of this specification in order to aid in comprehending the current disclosure. The pictures demonstrate exemplary implementations of the current disclosure and, along with the description, assist to clarify its fundamental ideas.

20 [0032] Fig.1 shows the Secure Spoofing Detection Platform - High-level system architecture showing the Verification Engine, AI Detection Module, Authentication Service, Database, and integration with external entities including STIR/SHAKEN Service, Carrier APIs, and Messaging Gateway.

25 [0033] Fig 2 shows the AI-Based Spoofing Detection System operates through a sequential pipeline of five integrated AI models.

[0034] Fig 3 shows the Distribution analysis of key features showing normal vs. attack traffic patterns.

[0035] Fig 4 shows the LSTM model training history showing loss convergence over 50 epochs.

[0036] Fig 5 shows the confusion matrix for the Behavioral Pattern Learning model showing classification performance.

[0037] Fig 6 shows the distribution of anomaly scores showing clear separation between normal and attack traffic.

5 [0038] Fig 7 shows the Temporal threat heatmap showing predicted attack probabilities over time.

[0039] Fig 8 shows the DQN agent reward history showing policy improvement during training.

10 [0040] Fig 9 shows the feature importance analysis showing contribution of each model to final detection.

[0041] Fig 10 shows the real-time detection timeline showing system response to attack events.

[0042] Fig 11 shows the confusion matrix for the final ensemble model showing overall system performance.

15 [0043] It should be noted that the figures are not drawn to scale, and the elements of similar structure and functions are generally represented by like reference numerals for illustrative purposes throughout the figures. It should be noted that the figures do not illustrate every aspect of the described embodiment and do not limit the scope of the present disclosure.

20 [0044] Other objects, advantages, and novel features of the invention will become apparent from the following detailed description of the present embodiment when taken in conjunction with the accompanying drawings.

DETAILED DESCRIPTION OF THE INVENTION

25 [0045] While the present invention is described herein by way of example using embodiments and illustrative drawings, those skilled in the art will recognize that the invention is not limited to the embodiments of drawing or drawings described and are not intended to represent the scale of the various components. Further, some components that may form a part of the invention may not be illustrated in certain figures, for ease of illustration, and such omissions do

not limit the embodiments outlined in any way. It should be understood that the drawings and the detailed description thereto are not intended to limit the invention to the particular form disclosed, but on the contrary, the invention is to cover all modifications, equivalents, and alternatives falling within the scope of the present invention as defined by the appended claim.

[0046] As used throughout this description, the word "may" is used in a permissive sense (i.e. meaning having the potential to), rather than the mandatory sense, (i.e. meaning must). Further, the words "a" or "an" mean "at least one" and the word "plurality" means "one or more" unless otherwise mentioned.

Furthermore, the terminology and phraseology used herein are solely used for descriptive purposes and should not be construed as limiting in scope. Language such as "including," "comprising," "having," "containing," or "involving," and variations thereof, is intended to be broad and encompass the subject matter listed thereafter, equivalents, and additional subject matter not recited, and is not intended to exclude other additives, components, integers, or steps. Likewise, the term "comprising" is considered synonymous with the terms "including" or "containing" for applicable legal purposes. Any discussion of documents, acts, materials, devices, articles, and the like are included in the specification solely for the purpose of providing a context for the present invention. It is not suggested or represented that any or all these matters form part of the prior art base or were common general knowledge in the field relevant to the present invention.

[0047] In this disclosure, whenever a composition or an element or a group of elements is preceded with the transitional phrase "comprising", it is understood that we also contemplate the same composition, element, or group of elements with transitional phrases "consisting of", "consisting", "selected from the group of consisting of", "including", or "is" preceding the recitation of the composition, element or group of elements and vice versa.

[0048] The present invention is described hereinafter by various embodiments with reference to the accompanying drawing, wherein reference numerals used in the accompanying drawing correspond to the like elements throughout the description. This invention may, however, be embodied in many different forms

and should not be construed as limited to the embodiment set forth herein. Rather, the embodiment is provided so that this disclosure will be thorough and complete and will fully convey the scope of the invention to those skilled in the art. In the following detailed description, numeric values and ranges are provided for various aspects of the implementations described. These values and ranges are to be treated as examples only and are not intended to limit the scope of the claims. In addition, several materials are identified as suitable for various facets of the implementations.

[0049] The present invention relates to an artificial intelligence-based multi-layer spoofing detection and response system configured to detect and mitigate spoofing attacks in real-time network environments. The invention integrates multiple machine learning and deep learning models to analyze network behavior, detect anomalies, predict potential threats, and dynamically respond to suspicious activities.

[0050] In an embodiment of the present invention, an AI-based multi-layer spoofing detection and response system for real-time network security, comprises: (i) a data ingestion module configured to receive network traffic data associated with a plurality of devices and to extract a feature vector comprising network-level features, device-level features, and behavioral features; (ii) a behavioral pattern learning engine comprising a bidirectional long short-term memory (BiLSTM) network configured to process temporal sequences of consecutive network events for each device and to generate a behavioral deviation score indicative of spoofing activity; (iii) a real-time anomaly detection engine with an autoencoder model configured to generate a reconstruction error from the feature vector, and an isolation forest model configured to generate a statistical outlier score for the feature vector; (iv) a predictive threat intelligence engine comprising a long short-term memory (LSTM) predictor configured to estimate a probability of an upcoming attack and an attack category based on historical temporal sequences; (v) an adaptive policy network comprising a deep Q-network (DQN) configured to select at least one detection-threshold adjustment and/or response action based on current network state metrics; and (vi) a multi-layer correlation

engine configured to aggregate outputs from the BiLSTM network, the anomaly detection engine, the predictive threat intelligence engine, and the DQN, to produce a spoofing decision and to trigger a response in real time.

[0051] In another embodiment of the present invention, the method for real-time spoofing detection in a network, comprises: (i) ingesting network traffic data and extracting a feature vector comprising network-level features, device-level features, and behavioral features; (ii) generating a behavioral deviation score by processing sequences of consecutive network events using a bidirectional LSTM network; (iii) generating an anomaly score by computing an autoencoder reconstruction error and an isolation forest outlier score for the feature vector; (iv) predicting an upcoming attack probability and attack category using an LSTM predictor based on historical sequences; (v) selecting, using a deep Q-network, at least one detection-threshold adjustment and/or response action based on current network state metrics; (vi) correlating outputs and using a multi-layer correlation engine to generate a spoofing decision; and (vii) executing a real-time response action based on the spoofing decision.

[0052] In an aspect of the present invention, the system ingests network traffic data containing 50+ features categorized into: i) Network-Level Features: Source/Destination IP addresses and ports, Protocol type (TCP, UDP, ICMP, etc.), Packet size and flow statistics, Connection duration and timing; ii) Device-Level Features: MAC addresses and device fingerprints, Operating system signatures, Historical behavior profiles, Trust scores; iii) Behavioral Features: Time-of-day patterns, Traffic volume patterns.

[0053] Behavioral Pattern Learning (LSTM)

[0054] In another aspect of the present invention, architecture: i) Input: Sequences of 20 network events (20×50 features); ii) Bidirectional LSTM with 64 units per direction; iii) Dropout regularization (0.3) for generalization; iv) Dense output layer with sigmoid activation.

[0055] Anomaly Detection (Autoencoder + Isolation Forest)

[0056] In another aspect of the present invention, autoencoder architecture: i) Encoder: $50 \rightarrow 32 \rightarrow 16 \rightarrow 8$ dimensions (compression); ii) Latent space: 8 dimensions; iii) Decoder: $8 \rightarrow 16 \rightarrow 32 \rightarrow 50$ dimensions (reconstruction); iv) Loss function: Mean Squared Error.

5 **[0057] Predictive Threat Intelligence**

[0058] In another aspect of the present invention, architecture: i) LSTM-based sequence predictor; ii) Input: Historical attack sequences (30 time steps); iii) Output: Attack probability for 5 attack categories.

[0059] Adaptive Self-Learning Policies (DQN)

10 **[0060]** In another aspect of the present invention, architecture: i) State space: Current network metrics (packet rate, anomaly rate, false positive rate); ii) Action space: 5 discrete actions (adjust thresholds, block, allow, flag, escalate); iii) Reward function: +1 for correct detection, -1 for false positive, -2 for missed attack.

15 **[0061]** In another aspect of the present invention, policy adaptation: the DQN continuously adapts detection thresholds based on: i) Current network traffic patterns, ii) Recent false positive/negative rates, iii) Attack frequency and severity, iv) Time-of-day patterns.

[0062] Multi-Layer Correlation Engine

20 **[0063]** In another aspect of the present invention, architecture: i) Random Forest Classifier with 100 estimators; ii) Input: Concatenated outputs from Features 1-4; iii) Cross-layer validation logic; iv) Confidence-weighted voting.

[0064] Dataset Description

Parameter	Value
Total Samples	100,000
Normal Traffic	85,000 (85%)
Attack Traffic	15,000 (15%)
Attack Types	5 (IP, MAC, ARP, DNS, DDoS)
Features	50+
Training Set	70%
Validation Set	15%

Test Set	15%
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[0065] Summary Performance Metrics

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Behavioral LSTM	97.2%	96.8%	95.4%	96.1%	0.987
Anomaly Detector	96.5%	95.2%	97.1%	96.1%	0.982
Threat Predictor	95.8%	94.6%	96.3%	95.4%	0.975
DQN Policy	94.2%	93.1%	95.8%	94.4%	0.968
Correlation Engine	98.9%	98.5%	99.2%	98.8%	0.995
Ensemble (Final)	98.7%	98.3%	99.1%	98.7%	0.993

[0066] Attack-Specific Detection Rates

Attack Type	Detection Rate	False Positive Rate
IP Spoofing	99.2%	0.8%
MAC Spoofing	98.5%	1.2%
ARP Spoofing	98.8%	1.0%
DNS Spoofing	97.9%	1.5%
DDoS	99.5%	0.5%

[0067] Comparison with Prior Art

System	Accuracy	False Positive Rate	Adaptive	Multi-Layer
Traditional IDS	85-90%	15-25%	No	No
ML-based (Single Model)	92-95%	8-12%	No	No
Deep Learning	94-96%	5-8%	Limited	No

(Prior Art)				
This Invention	98.7%	1.1%	Yes (DQN)	Yes (5 layers)

[0068] In another aspect of the present invention, the AI-Based Spoofing Detection System operates through a sequential pipeline of five integrated AI models: i) Behavioral Learning: The system continuously learns normal behavior patterns for each network device using Bidirectional LSTM networks that process temporal sequences of 20 consecutive network events. Deviations from learned patterns trigger behavioral alerts. ii) Anomaly Detection: A dual-model approach combines Autoencoder reconstruction error (measuring how well input patterns can be reconstructed from a compressed latent space) with Isolation Forest statistical outlier detection (measuring how many splits are needed to isolate a data point). iii) Threat Prediction: Historical attack patterns are analyzed using LSTM networks to predict the probability and type of upcoming attacks, enabling proactive defense measures. iv) Adaptive Policies: A Deep Q-Network learns optimal detection thresholds and response actions through continuous interaction with the network environment, balancing detection sensitivity against false alarm rates. v) Multi-Layer Correlation: The ensemble classifier aggregates outputs from all models, weighting each signal based on its reliability and cross-validating detections across multiple feature layers.

[0069] Further, the operations need not be performed in the disclosed order, although in some examples, an order may be preferred. Also, not all functions need to be performed to achieve the desired advantages of the disclosed system and method, and therefore not all functions are required.

[0070] Various modifications to these embodiments are apparent to those skilled in the art from the description and the accompanying drawings. The principles associated with the various embodiments described herein may be applied to other embodiments. Therefore, the description is not intended to be limited to the embodiments shown along with the accompanying drawings but is to be providing the broadest scope consistent with the principles and the novel and inventive features disclosed or suggested herein. Accordingly, the invention is

anticipated to hold on to all other such alternatives, modifications, and variations that fall within the scope of the present invention and appended claims.

CLAIMS

I/We Claim:

1. An AI-based multi-layer spoofing detection and response system for real-time network security, comprising:

- 5 (i) a data ingestion module configured to receive network traffic data associated with a plurality of devices and to extract a feature vector comprising network-level features, device-level features, and behavioral features;
- 10 (ii) a behavioral pattern learning engine comprising a bidirectional long short-term memory (BiLSTM) network configured to process temporal sequences of consecutive network events for each device and to generate a behavioral deviation score indicative of spoofing activity;
- 15 (iii) a real-time anomaly detection engine with an autoencoder model configured to generate a reconstruction error from the feature vector, and an isolation forest model configured to generate a statistical outlier score for the feature vector;
- 20 (iv) a predictive threat intelligence engine comprising a long short-term memory (LSTM) predictor configured to estimate a probability of an upcoming attack and an attack category based on historical temporal sequences;
- (v) an adaptive policy network comprising a deep Q-network (DQN) configured to select at least one detection-threshold adjustment and/or response action based on current network state metrics; and
- 25 (vi) a multi-layer correlation engine configured to aggregate outputs from the BiLSTM network, the anomaly detection engine, the predictive threat intelligence engine, and the DQN, to produce a spoofing decision and to trigger a response in real time.

2. The system as claimed in claim 1, wherein the network-level features comprise one or more of source IP address, destination IP address, source port, destination port, protocol type, packet size, flow statistics, connection duration, and timing attributes.
- 5 3. The system as claimed in claim 1, wherein the BiLSTM network processes sequences of twenty consecutive network events represented as a tensor of size $20 \times N$, where N is at least fifty features, and outputs the behavioral deviation score using a sigmoid-activated dense layer.
- 10 4. The system as claimed in claim 1, wherein the autoencoder model comprises an encoder performing dimensionality reduction from at least fifty dimensions to a latent representation of eight dimensions via intermediate layers, and a decoder configured to reconstruct the feature vector, and wherein the reconstruction error is computed using mean squared error.
- 15 5. The method for real-time spoofing detection in a network, as claimed in claim 1, comprising:
- (i) ingesting network traffic data and extracting a feature vector comprising network-level features, device-level features, and behavioral features;
 - 20 (ii) generating a behavioral deviation score by processing sequences of consecutive network events using a bidirectional LSTM network;
 - (iii) generating an anomaly score by computing an autoencoder reconstruction error and an isolation forest outlier score for the feature vector;
 - 25 (iv) predicting an upcoming attack probability and attack category using an LSTM predictor based on historical sequences;
 - (v) selecting, using a deep Q-network, at least one detection-threshold adjustment and/or response action based on current network state metrics;

(vi) correlating outputs and using a multi-layer correlation engine to generate a spoofing decision; and

(vii) executing a real-time response action based on the spoofing decision.

6. The method as claimed in claim 5, wherein the bidirectional LSTM processes
5 sequences of twenty consecutive network events each having at least fifty features.

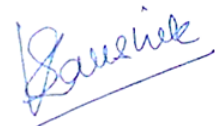
7. The method as claimed in claim 5, wherein the autoencoder reduces a feature dimension from about fifty to a latent space of eight dimensions and computes reconstruction error using mean squared error.

10 8. The system as claimed in claim 1, wherein the multi-layer correlation engine comprises a random forest classifier having about one hundred estimators and configured to receive concatenated outputs from the BiLSTM network, the autoencoder model, the isolation forest model, and the LSTM predictor.

15 9. The system as claimed in claim 1, wherein the multi-layer correlation engine applies confidence-weighted voting by assigning weights to model outputs based on reliability measures computed from cross-layer validation.

Dated this 19th day of March 2026

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ABSTRACT

“An AI-based multi-layer spoofing detection and response system for real-time network security”

The present invention relates to an artificial intelligence-based system for
5 detecting and responding to spoofing attacks in real-time network environments.
The system comprises a data ingestion module configured to collect network
traffic data from multiple devices and extract feature vectors including network-
level, device-level, and behavioral attributes. A behavioral pattern learning engine
employing a bidirectional long short-term memory (BiLSTM) network analyzes
10 temporal sequences of network events to generate a behavioral deviation score
indicative of spoofing activity. A real-time anomaly detection engine utilizes an
autoencoder model and an isolation forest model to identify abnormal patterns
within the extracted features. A predictive threat intelligence engine incorporating
a long short-term memory (LSTM) predictor estimates the probability and
15 category of potential upcoming attacks based on historical traffic sequences. An
adaptive policy network using a deep Q-network (DQN) dynamically adjusts
detection thresholds and selects appropriate response actions according to current
network conditions. A multi-layer correlation engine aggregates outputs from
multiple analytical layers to generate a spoofing decision and trigger automated
20 real-time mitigation actions. The invention provides an adaptive and intelligent
framework for accurate detection, prediction, and response to spoofing attacks,
thereby enhancing overall network security.

Fig 1 to 11.

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